

EXPERIMENT – 40

Implementation of Letter Frequency Attack on Monoalphabetic Substitution Cipher

CODE:

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

char freq[] = "ETAOINSHRDLCLUMWFGYPBVKJXQZ";

int main()
{
    char cipher[500], plain[500];
    int count[26] = {0};
    int order[26], i, j, t, top;

    printf("Enter ciphertext: ");
    fgets(cipher, sizeof(cipher), stdin);

    printf("Enter number of possible plaintexts: ");
    scanf("%d", &top);

    /* Count frequency of letters */
    for (i = 0; cipher[i]; i++)
        if (isalpha(cipher[i]))
            count[toupper(cipher[i]) - 'A']++;

    /* Sort letters by frequency */
```

```
for (i = 0; i < 26; i++)
```

```
    order[i] = i;
```

```
for (i = 0; i < 25; i++)
```

```
    for (j = i + 1; j < 26; j++)
```

```
        if (count[order[i]] < count[order[j]])
```

```
        {
```

```
            t = order[i];
```

```
            order[i] = order[j];
```

```
            order[j] = t;
```

```
        }
```

```
printf("\nTop %d possible plaintexts:\n", top);
```

```
for (t = 0; t < top && t < 26; t++)
```

```
{
```

```
    char map[26];
```

```
    /* Map frequent cipher letters to common English letters */
```

```
    for (i = 0; i < 26; i++)
```

```
        map[order[i]] = freq[(i + t) % 26];
```

```
    for (i = 0; cipher[i]; i++)
```

```
    {
```

```
        if (isalpha(cipher[i]))
```

```
            plain[i] = map[toupper(cipher[i]) - 'A'];
```

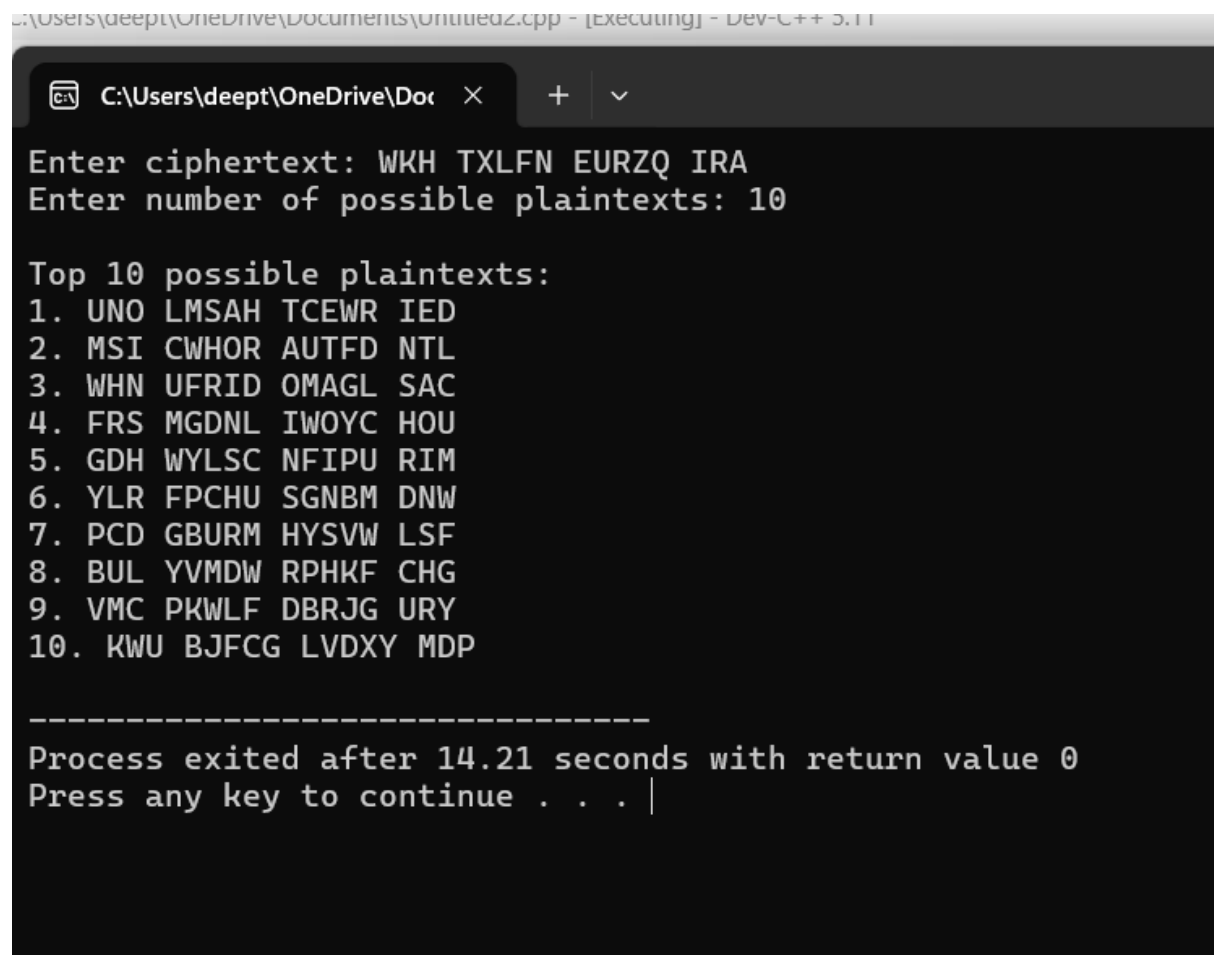
```
        else
```

```
            plain[i] = cipher[i];
```

```
    }
```

```
    plain[i] = '\0';  
    printf("%d. %s", t + 1, plain);  
}  
  
return 0;  
}
```

OUTPUT:



```
C:\Users\deep\OneDrive\Documents\Untitled2.cpp - [executing] - Dev-C++ 5.11  
Enter ciphertext: WKH TXLFN EURZQ IRA  
Enter number of possible plaintexts: 10  
  
Top 10 possible plaintexts:  
1. UNO LMSAH TCEWR IED  
2. MSI CWHOR AUTFD NTL  
3. WHN UFRID OMAGL SAC  
4. FRS MGDNL IWOYC HOU  
5. GDH WYLSC NFIPU RIM  
6. YLR FPCHU SGNBM DNW  
7. PCD GBURM HYSVW LSF  
8. BUL YVMDW RPHKF CHG  
9. VMC PKWLF DBRJG URY  
10. KWU BJFCG LVXDY MDP  
  
-----  
Process exited after 14.21 seconds with return value 0  
Press any key to continue . . . |
```